

Tables, Graphs, and Descriptive Statistics

QBA 5330 – Module 1, Part 3 (September 4)

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We know how to frame a decision, and where trustworthy data comes from.

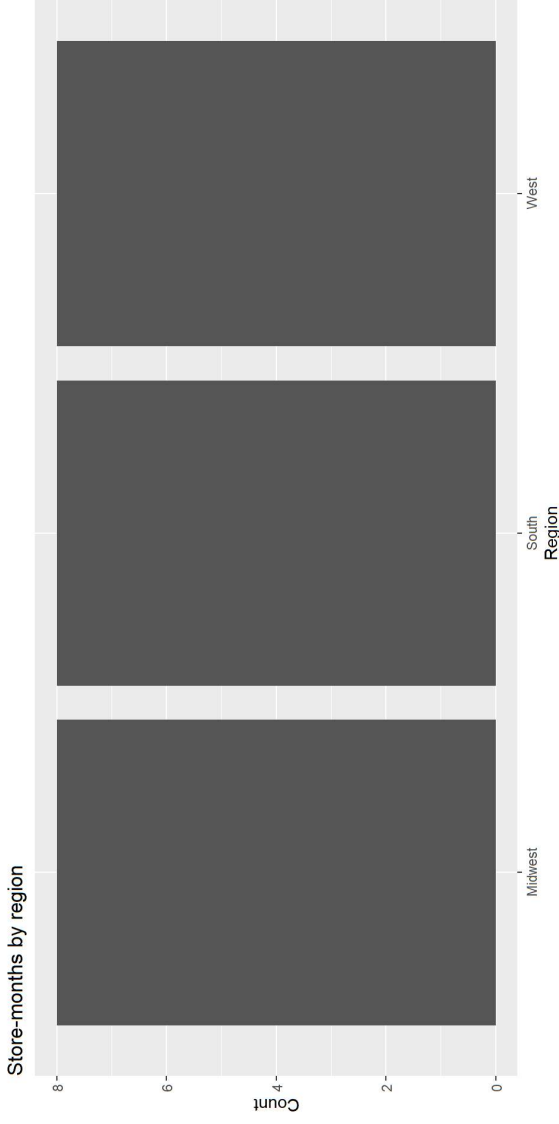
Now: how do we actually look at it?

Matching the Graph to the Variable

- **Categorical variable** → frequency table → **bar chart**
- **Quantitative variable** → binned counts → **histogram**

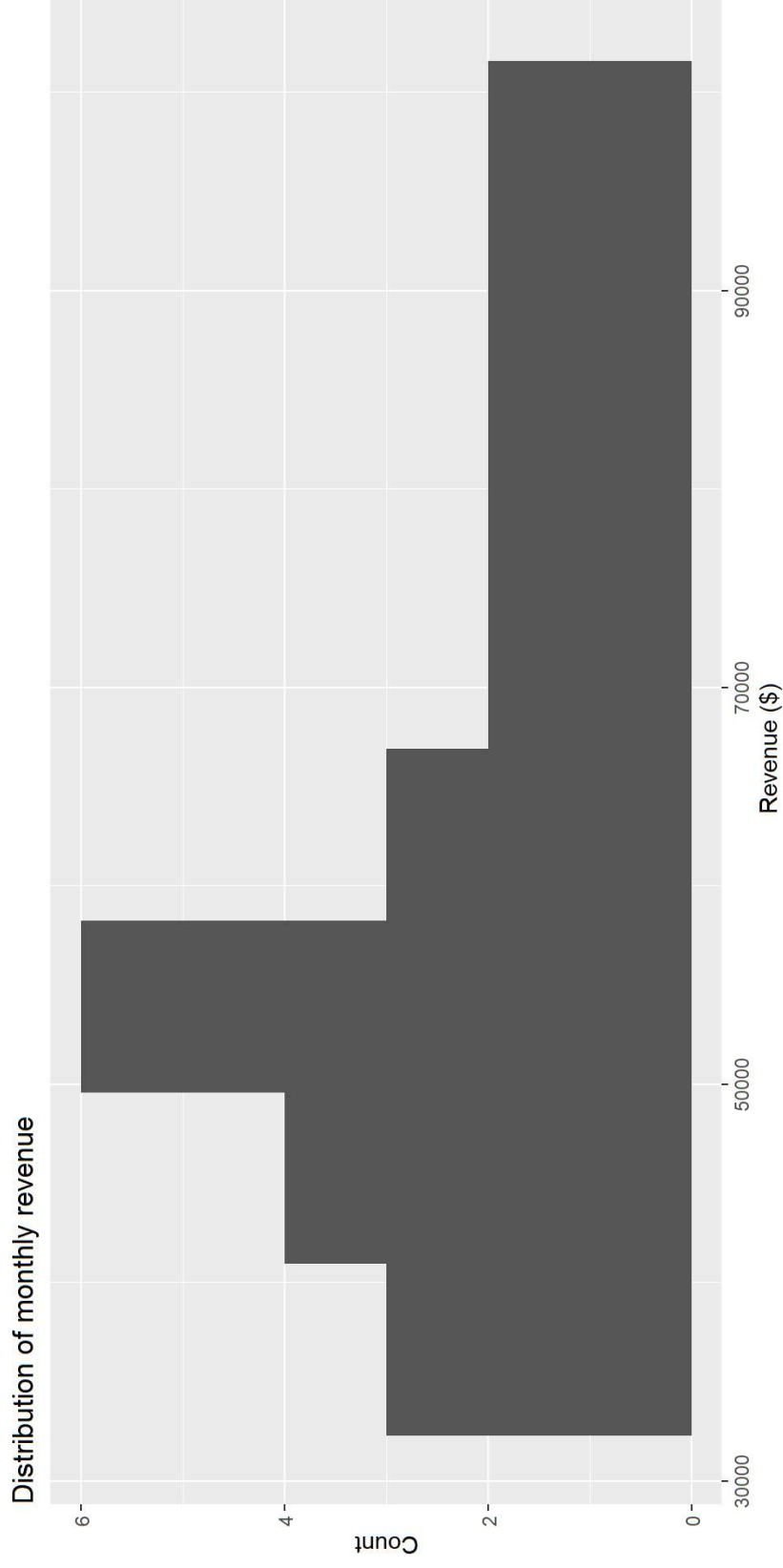
Bar Chart: **region** (Categorical)

```
1 library(tidyverse)
2 sales <- read_csv("store_sales.csv")
3
4 sales |>
5   count(region) |>
6   ggplot(aes(x = region, y = n)) +
7   geom_col() +
8   labs(title = "Store-months by region", x = "Region", y = "Count")
```



Histogram: revenue (Quantitative)

```
1 ggplot(sales, aes(x = revenue)) +  
2   geom_histogram(bins = 8) +  
3   labs(title = "Distribution of monthly revenue", x = "Revenue ($)", y = "Count")
```



A Common Mix-Up

Building a bar chart of a **quantitative** variable's raw values (one bar per row) instead of a histogram.

Center and Spread

- **Mean** — arithmetic average; sensitive to extreme values
- **Median** — middle value when sorted; resistant to extreme values
- **Standard deviation** — typical distance from the mean

```
1 sales |>
2   summarize(
3     mean_revenue = mean(revenue),
4     median_revenue = median(revenue),
5     sd_revenue = sd(revenue)
6   )
```

```
# A tibble: 1 × 3
```

	mean_revenue	median_revenue	sd_revenue
	<dbl>	<dbl>	<dbl>
1	61491.	57750	17808.

When the Mean Misleads

Add one hypothetical flagship store with unusually high revenue...

- **Mean** shifts noticeably
- **Median** barely moves

Means are pulled toward extreme values; medians are not.

Your Turn

Using `store_sales.csv`:

1. Build a frequency table for `region`
2. Build a histogram of `units_sold`
3. Compute the mean and median of `units_sold` – are they close together?